

Selecting Evidence Representations by Reader Does Not Pay: A Bounded Negative Result for Reader-Conditioned Compression

Anonymous ACL submission

Abstract

Post-retrieval compression is reader-dependent, and the field’s next step — condition the compressor on reader identity — is being actively pursued. We test it and find an inversion: given the *maximal* conditioning signal (20 donor readers’ measured outcomes on the very row being routed), selecting evidence by the most similar reader scores 57.6 while ignoring reader identity and averaging scores 60.5. Conditioning subtracts value at fixed information. We explain this with one hypothesis — the per-row routing signal is noise-dominated (per-row compression deltas of two readers correlate at only 0.38/0.16) — and confirm its three derived predictions on a 21-reader $\times$ 500-query $\times$ 3-representation panel of frozen, replayed compilation outputs (191k judged cells): realizable reader-conditioned policies gain ≤ 0.3 pp over reader-blind counterparts; similarity-weighted ensembles gain +0.25pp over blind averaging; and the cross-reader correlations replicate on held-out benchmarks (8 of 9 pre-registered cells; the single violation — a trained compressor in-domain, $r = 0.58$ — is the shared-signal regime that reader-blind routing captures by construction). The query-conditioned rung of the same policy ladder gains +2.5pp with no reader information. For selection among fixed representations, the conditioning variable that pays is the query, not the reader. We release the matrix and all analysis code.

1 Introduction

A fixed post-retrieval compressor affects readers asymmetrically: it rescues weak readers by removing noise they cannot filter and damages strong readers by dropping details they would have used (Panthi and Abdelfattah, 2026; Jeong et al., 2025; Xu et al., 2024). The natural method-side response is under way: condition the compression on the reader. FAVICOMP (Jung et al., 2025) shapes evidence toward the target model’s token distributions; the routing and capability-profiling literature (Ong

et al., 2024; Anonymous, 2026a,c) supplies ever-richer reader representations a compression policy could condition on; and our own prior work quantified the invitation — 37% of rows are mixed-label (the same evidence helps some readers and hurts others), and a reader-blind per-row oracle accesses only $\sim 70\%$ of routing headroom, implying the rest needs reader identity.

Here is what happens when that invitation is accepted under the most favorable conditions we can construct. Take 21 readers with frozen, replayed evidence (three representations per query, 191,277 judged cells), hold one reader out, and give the routing policy the strongest conditioning signal that can exist: the *measured outcomes* of the 20 other readers on the very row being routed — strictly more information than any deployable reader-profiler could ever extract. Then compare two policies. Ignoring reader identity (majority vote across donors) scores **60.5**. Conditioning on it (follow the single most similar donor, with similarity chosen with test-set hindsight) scores **57.6**. Conditioning on the reader does not merely fail to help; *at fixed information, it subtracts value*.

We propose one hypothesis that explains the inversion, derive three predictions from it, and confirm all three.

Hypothesis (noise-dominated routing signal).

The per-row, per-reader compression delta — which representation helps *this* reader on *this* row — is mostly noise around a shared query-level signal: per-row deltas of any two readers correlate at only 0.38 (structured—raw) and 0.16 (summary—raw). If true, cross-reader averaging *denoises* the signal while conditioning on any one reader’s identity or profile *re-injects* the noise. This forces:

- **P1.** No realizable reader-conditioned selection policy can materially beat its reader-blind counterpart, regardless of features or calibration budget. *Confirmed:* across per-reader best-fixed

(+0.2pp), per-(reader, question-type) with up to 200 labeled calibration rows (+0.0pp), and leave-family-out transfer (+0.25pp), no rung exceeds +0.3pp (§5, Table 1).

- **P2.** Inside the maximal-information ensemble regime, weighting donors by similarity to the held-out reader adds $\sim$ nothing over blind averaging, and concentrating on the most similar donor actively loses. *Confirmed:* +0.25pp and -2.9 pp respectively (§5, Table 2).
- **P3.** The low cross-reader correlation is a property of the routing problem, not of conversational-memory QA: it must replicate on other benchmarks in the same matrix. *Outcome — held in 8 of 9 cells, violated in one, and the violation is informative.* We pre-registered $r < 0.5$ before computing. Generic compression replicates everywhere (mean pairwise r : HotpotQA 0.39, MuSiQue 0.50, NQ 0.24, vs 0.16 on LongMemEval). The single violation is a *trained* compressor on its own training benchmark (RECOMP-style summary on HotpotQA: mean $r = 0.58$). High cross-reader correlation is the *shared-signal* regime: a trained compressor deletes content systematically, damaging all readers together — which reader-blind routing captures by construction. Both regimes exclude reader conditioning, for opposite reasons (§5).

The same ladder that kills reader conditioning locates where adaptivity does pay: a *query*-conditioned policy (question type $\rightarrow$ representation) gains +2.5pp reader-blind, against ≤ 0.3 pp for reader identity. Scope, stated up front: this bounds *selection among fixed representation styles* — the action space of the adaptive-compression literature’s routing variants. It does not bound generative reader-conditioned compression (white-box decoding-time shaping), which our black-box setting excludes.

The practical conclusions: (i) condition on the query, not the reader; (ii) reader-conditioned compression papers should report the reader-blind ablation under reader-family holdout — on our panel that ablation erases the conditioning gain entirely; (iii) multi-reader evaluation is not only a measurement obligation (as prior work argues) but the *optimal policy-learning input*, because averaging across readers is the denoiser.

2 Related Work

Adaptive compressors adapt ratio or query complexity, never reader identity: RECOMP (Xu et al., 2024), EXIT (Hwang et al., 2025), ECoRAG (Jeong et al., 2025), AdaComp (Zhang et al., 2024), ACC-RAG (Anonymous, 2025), PoC (Anonymous, 2026b), FAVICOMP (Jung et al., 2025) is reader-aware but generative and white-box — the case we explicitly do not bound. Routing work (Ong et al., 2024; Anonymous, 2026a,c) selects which model answers; we test the converse — selecting evidence for a fixed model — and find the per-row conditioning signal absent. Panthi and Abdelfattah (2026) establish reader-dependence and its measurement consequences; this paper tests the method direction that work implies, and corrects its headroom implication (§6).

3 Setup

Panel. 21 readers (7B–70B+, 11 families), complete uncapped triples on LongMemEval-S (500 queries): raw BM25 top-20, structured compilation, generic query-focused summary — compiled once, replayed unchanged (compile-once/replay-many). Binary correctness by an LLM judge under a fixed protocol; judge-disagreement flags (4.3% of cells) drive the noise controls. Five readers excluded for incomplete cells (availability-driven, not outcome-driven). Frozen evidence means every policy below — including oracles and ensembles no deployable system could run — is evaluated offline at zero marginal cost, which is what lets us *bound* the axis rather than report one policy’s failure.

Policies (10-fold CV over queries). Fixed: always-{raw, structured, summary}; per-reader best fixed. Query-conditioned: per-question-type (LongMemEval’s six native categories), reader-blind. Reader-conditioned: per-(reader, question-type); calibration sweep $K \in \{25\text{--}200\}$ rows/reader. Transfer (leave-reader-family-out, 100 reps), two regimes labeled explicitly: *ensemble bound* — donors’ measured outcomes on the test row (information ceiling; requires running 20 readers per query); *realizable* — donor similarity estimated from a 100-query calibration split only. Oracles: reader-blind per-row; per-(row, reader).

4 The Prize, Measured Honestly

Total oracle headroom over the best fixed policy: $65.8 - 52.8 = 13.0$ pp. The reader-identity

policy	conditions on	benchmark	Δ	style pair	mean r	% < 0.5
always raw / summary / struct.	—	46.7 / 50.0	LongMemEval (ref.)	summary—raw	0.16	100
per-reader best fixed	reader	53.0	LongMemEval (ref.)	struct.—raw	0.36	93
per-question-type	query	55.3	HotpotQA (cp.)	summary—raw	0.39	91
per-(reader, q-type)	reader $\times$ query	55.3	HotpotQA +0.0	tr.-abstr.—raw	0.58	25
$K=200$ calib. rows	reader $\times$ query	54.8	HotpotQA -0.5	tr.-extr.—raw	0.43	71
reader-blind per-row oracle	hindsight (query)	62.4	MuSiQue —	summary—raw	0.50	54
per-(row, reader) oracle	hind. (reader $\times$ query)	65.8	NQ +3.4	summary—raw	0.24	98
		NQ		tr.-abstr.—raw	0.38	73
		NQ		tr.-extr.—raw	0.40	83

Table 1: The policy ladder (accuracy, 10-fold CV; P1). Δ is versus the reader-blind counterpart.

policy for held-out reader	regime	acc
always structured	fixed	52.8
unweighted donor majority (blind)	ens. bound	60.5
similarity-weighted donor vote	ens. bound	60.8
single most-similar donor (hind.)	ens. bound	57.6

Table 2: Transfer to a held-out reader (leave-family-out; P2). All ensemble rows share identical (maximal) information; only the use of reader identity varies.

increment (per-(row, reader) oracle over reader-blind per-row oracle) is +3.4pp [95% CI 2.9–4.0] naively — but 4.3% of cells carry judge-disagreement flags, and dropping them shrinks the increment to +2.6pp, while injecting *random* label flips into flagged cells inflates it to +4.5pp. Hindsight per-reader oracles preferentially harvest label noise; published headroom estimates built this way (including our own prior $\sim 30\%$ -of-headroom figure) are biased upward by construction. So the prize is 20–26% of total headroom before asking whether any of it is reachable.

5 Results: The Predictions, Confirmed

Every realizable reader-conditioned rung is within ± 0.5 pp of its reader-blind counterpart, the best gains +0.3pp. Meanwhile query conditioning — same ladder, no reader information — gains +2.5pp. P1 holds (Table 1).

All three ensemble rows in Table 2 share the identical (maximal) information; only the use of reader identity varies. Weighting by similarity: +0.25pp (17/21 readers). Concentrating on the most similar reader — with hindsight — loses 2.9pp to blind averaging. P2 holds: at fixed information, conditioning re-injects exactly the noise averaging removes. (The 60.5 row is an information ceiling, not a deployable policy; it is not comparable to Table 1 and is reported to show conditioning fails even *above* deployability.)

The pre-registered threshold holds in 8 of 9 cells (Table 3); we report the violation rather than re-

Table 3: P3, cross-benchmark replication: mean pairwise per-row delta correlation across readers (pre-registered threshold $r < 0.5$). Readers per cell: 19 (HotpotQA/MuSiQue/NQ), 22–24 (LongMemEval).

vising it away. The violating cell is the trained abstractive compressor evaluated in-domain — the regime where compression decisions are most systematic. This maps the two regimes of the routing signal: **noise** (low r : per-row deltas are idiosyncratic, so conditioning on any reader re-injects noise — P1/P2) and **shared signal** (high r : systematic evidence deletion damages all readers together, so reader-blind routing captures it by construction). Reader conditioning pays only in a third regime — large, *stable*, *reader-specific* per-row structure — which no cell exhibits: where correlations are low the reader-specific residual is noise (split-half and transfer results, P1/P2), and where they are high the signal is not reader-specific at all.

What is real at the aggregate level — and why it doesn’t help. Per-reader action deltas are highly reliable (split-half $r = 0.845$ [0.737, 0.911]). The dominant pattern is a capability trend — representation effects shrink toward zero for the strongest readers — which is why a capability scalar exhausts the signal (+0.2pp, Table 1). Stable *exceptions* exist (equal-baseline pairs with opposite preferences: Phi-4 summary –3.6 vs Gemma-3-12B +7.2 at raw ≈ 47 ; OLMo-32B +18.2 vs MiMo-v2.5 –10.0), and they are precisely what per-reader policies try to exploit — recovering ≤ 0.3 pp, because the exceptions are aggregate-level while the routing decision is per-row, where noise dominates.

The query policy, operationally. Six native LongMemEval question types mapped to representations per training fold (temporal $\rightarrow$ structured; knowledge-update $\rightarrow$ raw; multi-session $\rightarrow$ summary), applied to held-out queries; smallest type cell ≈ 30 queries per test fold.

6 What This Corrects

Our prior “reader-blind oracle captures only $\sim 70\%$ of headroom” framing (Panthi and Abdelfattah, 2026) is technically true and practically misleading: the residual is partly judge noise harvested by hindsight oracles (§4) and otherwise unreachable by any realizable policy (§5). Corrected summary: on this action space, reader identity is worth $\leq 0.3\text{pp}$ to any policy we could construct, under both query holdout and reader-family holdout — and the ensemble bound shows this is a property of the signal, not of our policy class: any capability vector, familiarity score, or routing embedding is a compression of strictly less information than the donors’ realized outcomes that already fail.

7 Data and Code Release

This study builds on the **ragscale** interaction matrix released with Panthi and Abdelfattah (2026): we use that paper’s dataset as the substrate for all policy, oracle, and transfer analyses here, and extend it. We release: (1) the **new analysis layer** — policy-ladder, leave-family-out transfer, and oracle noise-correction code (every table regenerates in ~ 2 minutes on a laptop, offline); (2) the **cells added for this study** — the summary-policy cells, the additional reader completing the 21-reader panel, and the 2,000-call format-probe run with its byte-identity-tested harness; (3) the **P3 cross-benchmark mechanism analysis** over the matrix’s held-out benchmark cells. MIT license. The dataset contribution of the prior paper is not reclaimed here; this paper’s artifact is the analysis layer and the extension cells.

Limitations

Selection among three positively correlated styles from one pipeline; the wider six-style pilot chose one representation unanimously but at pilot scale. Primary benchmark is conversational-memory QA, English only; P3 extends the *mechanism* check to three other benchmarks but not the full policy ladder. No frontier-scale readers (strongest $\sim 57\%$ raw); the observed shrink-to-zero capability trend implies conditioning grows less useful with capability — extrapolation, stated as such. Judge noise handled by flag-dropping and injection sensitivity, not human re-adjudication. A 100-row probe of syntactic *format* conditioning (content byte-identical, 5 formats $\times$ 4 readers) was directionally null (swings 2–6pp; CIs include zero for

3 of 4 readers, one at [1.0, 12.0]) but is underpowered — reported as a pilot only. Generative reader-conditioned compression is outside the bounded action space.

References

- Anonymous. 2025. ACC-RAG: Adaptive context compression for retrieval-augmented generation. *arXiv preprint*.
- Anonymous. 2026a. InferenceDynamics: Efficient routing of language models via capability profiling. *arXiv preprint*.
- Anonymous. 2026b. PoC: Preference-oriented context compression for retrieval-augmented generation. *arXiv preprint*.
- Anonymous. 2026c. RouteProfile: Reader capability profiles for model routing. *arXiv preprint*.
- Taeho Hwang et al. 2025. EXIT: Context-aware extractive compression for enhancing retrieval-augmented generation. *arXiv preprint arXiv:2412.12559*.
- Yeonseok Jeong et al. 2025. ECoRAG: Evidentiality-guided compression for long context RAG. *arXiv preprint arXiv:2506.05167*.
- Dongwon Jung, Qin Zhang, Sungjin Zhou, et al. 2025. Familiarity-aware evidence compression for retrieval-augmented generation. In *Findings of the Association for Computational Linguistics: EMNLP 2025*.
- Isaac Ong, Amjad Almahairi, Vincent Wu, Wei-Lin Chiang, Tianhao Wu, Joseph E. Gonzalez, M. Waleed Kadous, and Ion Stoica. 2024. RouteLLM: Learning to route LLMs with preference data. *arXiv preprint arXiv:2406.18665*.
- Sugam Panthi and Rabab Abdelfattah. 2026. Fixed RAG compression collapses measured reader scaling. *arXiv preprint arXiv:2606.21807*.
- Fangyuan Xu, Weijia Shi, and Eunsol Choi. 2024. RECOMP: Improving retrieval-augmented LMs with context compression and selective augmentation. In *Proceedings of the International Conference on Learning Representations*.
- Qianchi Zhang et al. 2024. AdaComp: Extractive context compression with adaptive predictor for retrieval-augmented large language models. *arXiv preprint arXiv:2409.01579*.